higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

PLANT AND EQUIPMENT
NQF LEVEL 2

(12030052)

8 March 2019 (X-Paper)
09:00–12:00

This question paper consists of 5 pages.

NCV

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. ALL work you do NOT want to be marked must be clearly crossed out.
5. Write neatly and legibly.

NCV

QUESTION 1

1.1 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.1.1–1.1.5) in the ANSWER BOOK.

1.1.1 Using hand powered tool is NOT an example of mechanism.

1.1.2 Tools and equipment that are not in a good working condition will not delay the progress and quality of work.

1.1.3 Mechanism is when machines are used by people to do physical work.

1.1.4 A concrete batch plant has all the equipment that we use to make concrete at a large construction site.

1.1.5 During the iron age, it was possible to produce stronger, cheaper tools with iron than with bronze.

(5 × 1) **[5]**

QUESTION 2

2.1 State FIVE advantages of wearing PPE while working on a construction site. (5)

2.2 Give the abbreviation 'OHSA' in full. (1)

2.3 Explain TWO aspects of OHSA that are important to construction workers. (4)

2.4 Housekeeping refers to the daily routine that must be conducted in order for the workplace to be a safe and productive environment.

Explain FIVE consequences of poor housekeeping. (5)
[15]

QUESTION 3

3.1 Explain THREE different levels of first aid priorities. (6)

3.2 State the FIVE basic rules to be followed when giving first aid, in order to prevent the spreading of infections from an injured person. (5)

3.3 List FOUR symptoms of *shock*. (4)
[15]

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QUESTION 4

4.1 Name ONE tool that falls under general tools that can be used by all the trades' man. (1)

4.2 Give the use of the following tools:

4.2.1 Spirit level

4.2.2 Plumb bob

4.2.3 Caulking gun

4.2.4 Set square

4.2.5 Screwdrivers

(5 × 1) (5)

4.3 It is said that working with a blunt tool is more dangerous than working with a sharp tool.

State SIX consequences of working with a blunt tool. (6)

4.4 In order to conduct a specific job in a specific trade, it is compulsory to use the correct tools for the trade.

List THREE tools that will be found in the toolbox of the following trades:

4.4.1 Plumbing

4.4.2 Carpentry

4.4.3 Bricklaying

4.4.4 Tiling

(4 × 3) (12)

4.5 Explain how you would mark out or set out with a hexagon steel plate to conform to specifications with a reasonable tolerance. (3)

4.6 List THREE main types of chisels. (3)

[30]

QUESTION 5

5.1 Name the TWO main types of ladders that can be used. (2)

5.2 Explain the dangers of over-reaching when using a ladder. (4)

5.3 List FOUR parts that should be inspected on a ladder for safety. (4)

[10]

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QUESTION 6

6.1 A scaffold should be inspected by a trained person to identify faults and non-compliances.

List FIVE training modules an inspector of scaffolding should have. (5)

6.2 Explain the correct use of wheeled mobile towers. (5)

6.3 In order to make sure that the scaffolding is stable, the scaffolding erector must ensure the correct height to base ratio.

Advise the scaffolding erector on the correct height to base ratio to be used. (2)

6.4 State the most important factor to consider when installing a ladder in a mobile tower. (2)

6.5 Explain in point form FIVE safety rules that workers must adhere to when working on trestle scaffolding. (5)

6.6 Explain FIVE requirements for performing the application of bracing and requirements for plan bracing. (5)

6.7 Write the full expansion of the abbreviation *SWL* as used in plant and equipment. (1)

[25]

TOTAL: 100

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